

Final Progress Report

Model Overview

I built a U-Net-style convolutional autoencoder for audio denoising, designed to reconstruct clean audio from degraded MP3 inputs. The model processes log-scaled magnitude spectrograms (from STFT), learning to reverse compression artifacts and restore lost detail. The final model uses encoder/decoder blocks and uses log-scaled MSE to try and match human perception.

Dataset & Preprocessing

*Dataset: about 4,500 pairs of clean and degraded audio clips, each 5 seconds long, sampled at 44.1 kHz. Degraded clips are produced via MP3 compression using ffmpeg.

*Preprocessing:

- STFT with 'n_fft=1024', 'hop_length=256', 'win_length=1024'
- Extract magnitude and phase
- Feed log-scaled magnitude to the model: 'log1p(mag)'
- Predict the denoised log-magnitude; recombine with original phase during

inverse STFT

- Overlap-add using a squared Hann window

Training Progress

Model	Train Loss	Train MAE	Val Loss	Val MAE	Params
Model A	0.0035	0.0431	0.0302	0.1220	738,561
Model B	0.0021	0.0309	0.0174	0.1027	924,161
Model C	0.0015	0.0318	0.0115	0.1021	1,514,753

Challenges & Solutions

*STFT/iSTFT mismatch artifacts

- Problem: Reconstructed audio was too quiet or noisy.
- Solution: Switched from WAV to MP3 output to avoid 16-bit clipping; changed to squared-Hann overlap-add and applied normalization safeguards.

*Output shape mismatches

- Problem: Skip-connections broke due to off-by-one shape mismatches after upsampling.
- Solution: Explicit 'Cropping2D' layers ensured dimensional consistency between encoder and decoder.

*Checkpoint corruption

- Problem: Crashes during weight saving led to corrupted files.
- Solution: Atomically saved to a '.tmp' file and renamed after success ('os.replace'), making training crash-safe.

Validation Performance

Final production model: Model C

- Validation Loss (MSE): 0.0115
- Validation MAE: 0.1021
- Subjective quality: not good :(numerically it performed the best but it actually sounds worse than the degraded input audio.

Next Steps

1. Add perceptual loss: supplement MSE with multi-scale STFT or mel-spectrogram loss.
2. Phase enhancement: apply Griffin-Lim or a learned phase predictor to further improve realism.
3. Frequency band conditioning: train the model to only predict above 8 kHz, leaving lower frequencies untouched.